Hair Fix Salon Functional Requirements Document

Online Appointment Booking Platform Revised Draft

Item	Information
Prepared by	Robert Martinez-Olivera and Yaswanth Gosukonda
Course and section	DATABASE SYS IMPLEMT 01:198:437:01
Date	September 28, 2026
Version	2.1
Status	Revised draft incorporating instructor feedback

Document Purpose

This document defines the functional and data requirements for the Hair Fix Salon appointment platform. It establishes the workflows, permissions, validation rules, audit requirements, reports, and test evidence that will guide the ERD, normalized schema, MySQL implementation, and project demonstration.

Version Two Decisions

- Version 1.0 of the platform supports one seeded salon location. Customers do not select a location, and administrators do not maintain multiple locations in this release.
- Each account has one primary role: Customer, Staff, or Administrator.
- Rescheduling updates the existing appointment and preserves prior values in AppointmentHistory.
- An appointment may have multiple non-sensitive payment transactions, but the database does not process payment cards.
- Only assigned staff members and administrators may mark an appointment No-Show.

1 Project Identification

Item	Project information
Project title	Hair Fix Salon Online Appointment Booking Platform
Prepared by	Robert Martinez-Olivera and Yaswanth Gosukonda
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2 Business Problem and Project Purpose

Hair Fix Salon needs one reliable database for customer accounts, services, staff qualifications, work schedules, appointments, appointment history, and payment records. Phone calls, paper calendars, messages, and separate spreadsheets can create double-bookings, incomplete history, inconsistent pricing, missed schedule changes, and reports that cannot be reproduced.

The project will implement a relational database that supports controlled appointment booking and operational reporting. Customers will view active services and open times, then book or manage their own eligible appointments. Staff will view assigned work, update permitted appointment statuses, and record payments. Administrators will maintain accounts, services, staff assignments, schedules, appointments, and reports.

3 Scope

In scope	Out of scope
Customer registration and authentication data	Production website or mobile application
Customer and staff profiles	Plaintext passwords or sensitive payment-card data
One seeded salon location	Customer location selection or multi-location operations
Services, current prices, and durations	Payroll, commissions, tips, and benefits
Staff qualifications and schedules	Inventory and retail product sales
Availability, booking, rescheduling, and cancellation	Automated email or SMS delivery
Status transitions and immutable history	Medical records or insurance claims
Payment transaction records and operational reports	Real payment-gateway integration

Location scope The database contains one Location row as seed and reference data. The system assigns that location automatically to staff schedules and appointments. Multiple locations may be added in a later version, but no version 1.0 workflow depends on location selection, filtering, or maintenance.

4 Stakeholders and User Roles

Role	Responsibilities	Database needs
Customer	Maintains a personal account and manages eligible appointments.	Active services, open times, personal appointments, and personal transaction history.
Staff member	Performs assigned services and updates assigned appointments.	Own schedule, assigned appointment details, status controls, and payment recording.
Administrator	Maintains operational records and oversees protected actions.	All accounts, services, staff assignments, schedules, appointments, audit data, and reports.

Role	Responsibilities	Database needs
Salon manager or owner	Reviews operations and may use an Administrator account.	Revenue, appointment volume, service popularity, workload, cancellations, and no-shows.

Account model UserAccount stores identity, login, name, security, and one primary role. Customer and Staff store only role-specific profile data. An Administrator does not require a Staff profile in version 1.0. A person who needs a different role uses a separate demonstration account.

5 Role and Entitlement Matrix

Protected action	Customer	Staff	Administrator
View active services and open times	Yes	Yes	Yes
Create or edit own profile	Yes	Yes	Yes
Create an appointment	Own only	No	For customer support
Reschedule or cancel an appointment	Own eligible appointment	Assigned appointment when authorized	Any eligible appointment
View appointments	Own only	Assigned only	All
Change appointment status	Cancel own eligible appointment	Assigned appointment through approved transitions	Any appointment through approved transitions
Assign No-Show	No	Assigned appointment	Yes
Maintain services and staff-service assignments	No	No	Yes
Maintain recurring schedules and blocked periods	No	View own	Yes
Create payment transactions	No	Assigned appointment	Yes
View payment transactions	Own appointments	Assigned appointments	All
View AppointmentHistory	Own appointment events	Assigned appointment events	All
Update or delete AppointmentHistory	No	No	No ordinary operation
Run management reports	No	Own daily schedule only	Yes

6 Workflows and Use Cases

The following use cases define the controlled operations that the database must support. Each successful operation creates the stated records, and each rejected operation leaves no partial records.

Use Case 01 Customer Books an Appointment

Element	Details
Identifier	UC-01
Primary actor	Customer
Preconditions	The customer account is active. The selected service and staff member are active. The staff member is qualified for the service.
Trigger	The customer selects an active service and future date, then optionally selects a staff member.

Element	Details
Main flow	1. The system retrieves qualified staff members and available start times. 2. The customer selects an available start time. 3. The system validates the future date, recurring schedule, date-specific exceptions, blocked time, staff qualification, and appointment overlap. 4. The system stores the appointment with Scheduled status, BookedPrice, and BookedDurationMinutes. 5. The system creates an AppointmentHistory event for the booking. 6. The transaction commits and the system returns the appointment confirmation.
Alternate or error outcomes	If the time becomes unavailable or any validation fails, the system rejects the request and rolls back the appointment and history changes.
Postconditions	One valid Appointment row and one immutable AppointmentHistory row exist.

Use Case 02 Customer Reschedules or Cancels an Appointment

Element	Details
Identifier	UC-02
Primary actor	Customer
Preconditions	The customer owns the appointment. The appointment is future, has Scheduled or Confirmed status, and is at least 24 hours away.
Trigger	The customer selects reschedule or cancel for an eligible appointment.
Main flow	1. The system verifies ownership, status, future time, and the 24-hour cutoff. 2. For rescheduling, the customer selects a new open time and the system repeats all booking validations. 3. The system updates the same Appointment row for a reschedule or changes its status to Cancelled for a cancellation. 4. The system records the previous and new values, actor, event type, timestamp, and reason in AppointmentHistory. 5. Existing payment transactions remain linked to the same AppointmentID and are not automatically changed. 6. The transaction commits and the system returns the updated appointment.
Alternate or error outcomes	If the appointment is ineligible, inside the cutoff, or the new time conflicts, the operation is rejected and the existing appointment remains unchanged.
Postconditions	The appointment reflects the accepted change, an immutable history event preserves the prior values, and existing payments remain attached to the appointment.

Use Case 03 Staff Updates Appointment Status

Element	Details
Identifier	UC-03
Primary actor	Assigned staff member or Administrator
Preconditions	The actor has an active authorized account and the requested change is an approved transition.
Trigger	The actor selects a new status for an appointment.
Main flow	1. The system verifies the actor role and assignment. 2. The system checks the status-transition matrix. 3. The actor enters a reason when cancelling or assigning No-Show. 4. The system updates the Appointment status. 5. The system appends a history event with the previous status, new status, actor, time, and reason.
Alternate or error outcomes	An unauthorized actor or invalid transition is rejected. A customer cannot assign No-Show. A final status cannot return to an active status through the ordinary workflow.

Element	Details
Postconditions	The Appointment has the approved status and a corresponding immutable history event.

Use Case 04 Administrator Maintains Staff Availability

Element	Details
Identifier	UC-04
Primary actor	Administrator
Preconditions	The administrator account and target staff member are active.
Trigger	The administrator adds or changes recurring availability, a date-specific override, or a blocked period.
Main flow	1. The system validates the schedule type, date or day of week, start time, and end time. 2. The system rejects an invalid time range or an impermissible conflicting schedule record. 3. The system saves the schedule record. 4. Availability calculation applies blocked periods first, then date-specific availability, then recurring availability.
Alternate or error outcomes	If the record is invalid or conflicts with the schedule policy, no schedule change is saved. Existing appointments are not silently deleted or moved.
Postconditions	The valid schedule record is available to future availability queries.

Use Case 05 Staff Records a Payment Transaction

Element	Details
Identifier	UC-05
Primary actor	Assigned staff member or Administrator
Preconditions	The appointment exists in any allowed appointment status and the actor is authorized. No sensitive card data is entered.
Trigger	The actor records a charge or refund associated with an appointment.
Main flow	1. The system validates the appointment, actor, positive amount, transaction type, payment method, payment status, and optional unique reference. 2. The system records the payment transaction and the user who recorded it. 3. The system returns the saved transaction.
Alternate or error outcomes	A non-positive amount, duplicate supplied reference, invalid value, or unauthorized customer request is rejected.
Postconditions	One valid non-sensitive PaymentTransaction row exists. Only succeeded transactions affect revenue reports.

7 Functional Requirements

ID	Functional requirement	Priority	Related data
FR-01	The system shall store one account for each registered user with a unique email, password hash, name, one primary role, account status, and creation and update timestamps.	High	UserAccount, Role
FR-02	When an active account is authenticated, the system shall permit only the functions assigned to that account's role.	High	UserAccount, Role

ID	Functional requirement	Priority	Related data
FR-03	A customer shall be able to view and update that customer's profile, including a required phone number and an Email or Phone contact preference, without changing protected role or security fields.	High	Customer, UserAccount
FR-04	An administrator shall be able to maintain staff profiles, active status, and the services each staff member is qualified to perform.	High	Staff, StaffService, Service
FR-05	An administrator shall be able to maintain service name, description, current price, duration, and active status.	High	Service
FR-06	An administrator shall be able to maintain recurring staff availability, date-specific availability, and blocked periods. Blocked periods shall override date-specific and recurring availability.	High	StaffSchedule, Staff
FR-07	When a customer selects an active service, future date, and optionally a staff member, the system shall return only start times within an eligible schedule, long enough for the service, not blocked, and not overlapping an existing non-cancelled appointment.	High	Appointment, StaffSchedule, StaffService, Service
FR-08	A customer with an active account shall be able to create an appointment only for a future time, active service, active qualified staff member, and seeded active location. The system shall save the booked price and duration and create a history event.	High	Appointment, AppointmentHistory
FR-09	Before inserting or rescheduling an appointment, the controlled booking operation shall reject the request if an existing non-cancelled appointment for the same staff member overlaps the requested interval.	High	Appointment
FR-10	A customer may cancel or reschedule only that customer's own future Scheduled or Confirmed appointment at least 24 hours before its start time. The system shall preserve prior values, actor, timestamp, reason, and event type in AppointmentHistory.	High	Appointment, AppointmentHistory
FR-11	An assigned staff member or administrator shall be able to change appointment status only through approved transitions. Each successful change shall create an immutable history event.	High	Appointment, AppointmentHistory
FR-12	Only an assigned staff member or administrator shall be able to assign No-Show, and the actor shall record a reason or explanatory note.	High	Appointment, UserAccount
FR-13	An assigned staff member or administrator shall be able to record one or more non-sensitive payment transactions for any existing appointment status, including Cancelled or No-Show. Each transaction shall contain a positive amount, allowed type, method, status, timestamp, recorder, and unique reference when supplied.	High	PaymentTransaction, Appointment
FR-14	A customer shall be able to view only that customer's appointments and transactions. Staff may view only assigned appointments. Administrators may view all records.	High	Appointment, PaymentTransaction
FR-15	An administrator shall be able to maintain user status, staff records, services, staff-service assignments, schedules, and appointments while preserving protected history.	High	Operational tables
FR-16	The system shall provide a daily appointment schedule filtered by date, staff member, and status, ordered by appointment start time.	Medium	Appointment, Staff, Service
FR-17	An administrator shall be able to run revenue, appointment-volume, popular-service, staff-workload, cancellation, and no-show reports for a selected date range.	Medium	Appointment, PaymentTransaction, Service, Staff
FR-18	When an appointment is created, the system shall store BookedPrice and BookedDurationMinutes so later service edits do not alter historical appointments or reports.	High	Appointment, Service
FR-19	The system shall reject any status change not allowed by the documented transition matrix, including a change from Completed back to Scheduled.	High	Appointment, AppointmentStatus
FR-20	Availability validation, appointment insertion or rescheduling, and history insertion shall execute in one transaction. If any step fails, all steps shall roll back.	High	Appointment, AppointmentHistory
FR-21	When an appointment is cancelled or marked No-Show, the actor shall record an approved reason or explanatory note.	High	AppointmentHistory
FR-22	The system shall reject a staff schedule or blocked-time record with an invalid time range or impermissible conflicting schedule condition.	High	StaffSchedule
FR-23	When a service, staff member, or seeded location becomes inactive, the system shall prevent new bookings that use the inactive record while preserving existing appointments and history.	High	Service, Staff, Location, Appointment

ID	Functional requirement	Priority	Related data
FR-24	The system shall preserve AppointmentHistory as append-only audit data and prohibit ordinary update or delete operations by customers, staff, and application administrators.	High	AppointmentHistory
FR-25	Only staff and administrators may create payment transactions. Customers may view transactions for their own appointments but may not create, update, or delete them.	High	PaymentTransaction, Role

8 Data Requirements and Definitions

Entity	Core attributes	Identifier	Definition
Role	RoleID, RoleName, Description	RoleID	Seed Customer, Staff, and Administrator roles.
UserAccount	Email, PasswordHash, FirstName, LastName, RoleID, AccountStatus, CreatedAt, LastUpdatedAt	UserID	Email is unique. Status values are Active, Inactive, or Locked. One primary role per account.
Customer	UserID, Phone, ContactPreference, ProfileStatus	CustomerID	UserID is a unique foreign key. Phone is required. ContactPreference is Email or Phone. Names remain in UserAccount.
Staff	UserID, Phone, Specialty, ActiveFlag, HireDate, LocationID	StaffID	UserID is a unique foreign key. Names remain in UserAccount.
Location	SalonName, Address, Phone, TimeZone, ActiveFlag	LocationID	Version 1.0 contains one seeded Hair Fix Salon row.
Service	ServiceName, Description, CurrentPrice, DurationMinutes, ActiveFlag	ServiceID	Current values apply only to new bookings. Active service name should be unique.
StaffService	StaffID, ServiceID, IsActive	StaffID and ServiceID	Identifies services a staff member is qualified to perform.
StaffSchedule	StaffID, ScheduleType, DayOfWeek or ScheduleDate, StartTime, EndTime, EffectiveStartDate, EffectiveEndDate, Reason	ScheduleID	ScheduleType is RecurringAvailability, DateAvailability, or BlockedTime.
AppointmentStatus	StatusName and IsFinal	StatusID	Seed Scheduled, Confirmed, In Progress, Completed, Cancelled, and No-Show.
Appointment	CustomerID, StaffID, ServiceID, LocationID, StartDateTime, EndDateTime, BookedPrice, BookedDurationMinutes, StatusID, CreatedAt, UpdatedAt, CancelledAt, Notes	AppointmentID	One service per appointment. Price and duration are historical snapshots.
AppointmentHistory	AppointmentID, EventType, ChangedByUserID, ChangedAt, previous and new status, previous and new start or end time, previous and new staff or service, Reason, RequestID	HistoryID	Append-only audit data. RequestID links all changes from one controlled operation.
PaymentMethod	MethodName and ActiveFlag	PaymentMethodID	Seed allowed non-sensitive methods such as Cash, Card Record, or Other.
PaymentTransaction	AppointmentID, Amount, TransactionType, PaymentMethodID, PaymentStatus, TransactionDate, RecordedByUserID, CreatedAt, ExternalReference	TransactionID	Type is Charge or Refund. Status is Pending, Succeeded, Failed, or Voided. TransactionDate is distinct from CreatedAt.

Security boundary The database will not store plaintext passwords, complete card numbers, card security codes, or medical information. PaymentTransaction stores only information needed to demonstrate transaction history and reports.

9 Business Rules and Validation Rules

ID	Business rule	Planned enforcement
BR-01	Each account must have a unique UserID and case-insensitive unique email.	PRIMARY KEY and UNIQUE
BR-02	Each UserAccount must reference exactly one valid primary role.	NOT NULL foreign key

ID	Business rule	Planned enforcement
BR-03	Plaintext passwords and sensitive payment-card values must not be stored.	Schema and controlled input
BR-04	A service must have CurrentPrice at least zero and DurationMinutes greater than zero.	CHECK constraints
BR-05	Only active staff with an active StaffService assignment may be booked for an active service.	Controlled booking query
BR-06	A new appointment must be future-dated and reference one active customer, qualified staff member, active service, and the seeded active location.	Foreign keys and procedure
BR-07	Appointment EndDateTime must equal StartDateTime plus BookedDurationMinutes and must be later than StartDateTime.	Procedure and CHECK where supported
BR-08	A conflict exists for the same staff member when NewStart < ExistingEnd and NewEnd > ExistingStart and the existing appointment status is not Cancelled.	Controlled procedure or trigger
BR-09	Customer rescheduling updates the same Appointment row and is allowed only for the customer's own future Scheduled or Confirmed appointment.	Ownership and status validation
BR-10	A customer cancellation or reschedule must occur at least 24 hours before StartDateTime. Staff or administrators handle later exceptions.	Cutoff validation
BR-11	Appointment status must reference an approved AppointmentStatus value.	Lookup foreign key
BR-12	Status changes must follow the transition matrix and must create AppointmentHistory.	Controlled procedure
BR-13	Booking, rescheduling, cancellation, and status operations must create one or more immutable history events.	Transaction and restricted privileges
BR-14	Cancelled and No-Show outcomes require a reason or explanatory note.	NOT NULL validation in procedure
BR-15	BlockedTime overrides DateAvailability, and DateAvailability overrides RecurringAvailability for the same staff member and date.	Availability query precedence
BR-16	Schedule EndTime must be later than StartTime, and invalid or impermissibly conflicting records must be rejected.	CHECK and controlled procedure
BR-17	Payment Amount must be greater than zero. TransactionType, PaymentStatus, and PaymentMethod must use allowed values.	CHECK and lookup foreign keys
BR-18	Revenue equals succeeded Charge amounts minus succeeded Refund amounts. Pending, Failed, and Voided transactions do not count.	Report query
BR-19	Only assigned staff or administrators may create payment transactions. Customers have read-only access to their own transactions.	Role and ownership validation
BR-20	Deactivation prevents new use of a service, staff member, staff-service assignment, or location but does not delete historical records.	Active-status validation and foreign keys
BR-21	AppointmentHistory may be inserted only by controlled operations and may not be updated or deleted by ordinary application roles.	Database privileges and optional trigger
BR-22	Appointment and schedule times are interpreted in the seeded salon's US Eastern time zone.	Location reference data
BR-23	Each appointment contains one service in version 1.0.	NOT NULL ServiceID foreign key
BR-24	Availability validation, appointment change, and history insertion must commit or roll back as one transaction.	Database transaction
BR-25	Customer Phone is required, and ContactPreference must be Email or Phone.	NOT NULL and CHECK or lookup
BR-26	Rescheduling retains the same AppointmentID, so existing payment transactions remain linked and are not automatically modified or transferred.	Foreign key and controlled procedure
BR-27	Authorized staff or administrators may record Charge or Refund transactions for any appointment status, including Cancelled and No-Show. A payment transaction does not change appointment status.	Role validation and procedure

10 Appointment Status Transition Matrix

From status	Allowed next status	Authorized actor	Condition or result
Scheduled	Confirmed	Assigned staff or Administrator	Appointment has been accepted for service.
Scheduled	Cancelled	Customer before cutoff, assigned staff, or Administrator	Cancellation reason required.
Confirmed	In Progress	Assigned staff or Administrator	Service begins.
Confirmed	Cancelled	Customer before cutoff, assigned staff, or Administrator	Cancellation reason required.
Confirmed	No-Show	Assigned staff or Administrator	Reason or note required. Customer is prohibited.
In Progress	Completed	Assigned staff or Administrator	Service is finished.

Final statuses Completed, Cancelled, and No-Show are final in the ordinary workflow. A correction requires a controlled administrative process that appends a corrective history event rather than editing or deleting an existing history row.

11 Reports and Operational Query Requirements

ID	Report	Primary filters	Required result
RQ-01	Daily appointment schedule	Date, staff, and status	Start time, customer, service, staff, booked duration, and status; ordered by start time.
RQ-02	Appointments by status	Status and date range	Appointment, customer, staff, service, date, and status.
RQ-03	Customer appointment history	Customer	Appointment, date, service, staff, booked price, status, and transaction summary; newest first.
RQ-04	Monthly revenue summary	Transaction date range	Succeeded charges minus succeeded refunds, transaction count, and average succeeded charge.
RQ-05	Popular services report	Appointment date range	Service, appointment count, completed count, and revenue using BookedPrice and succeeded payments.
RQ-06	Staff workload report	Appointment date range and staff	Staff, appointment count, completed count, and booked service minutes.
RQ-07	Cancellation and no-show report	Outcome and date range	Appointment, customer, service, staff, final status, reason, actor, and change time.

Operational Query and Performance Requirements

Query pattern	Likely index direction	Reason
Staff calendar and conflict validation	Appointment by StaffID, StartDateTime, and Status	Supports daily schedules and overlap checks.
Customer history newest first	Appointment by CustomerID and StartDateTime	Supports ownership filtering and ordered history.
Revenue by date and status	PaymentTransaction by TransactionDate and PaymentStatus	Supports succeeded charge and refund reporting.
Service popularity	Appointment by ServiceID, StartDateTime, and Status	Supports grouped service counts over a date range.
Availability calculation	StaffSchedule by StaffID, ScheduleDate or DayOfWeek, and StartTime	Supports recurring, date-specific, and blocked-time lookup.
Appointment audit trail	AppointmentHistory by AppointmentID and ChangedAt	Supports chronological history review.

Index scope The final index names and column order belong in the technical design document and schema script. The team will compare query plans before and after indexing and retain indexes supported by the demonstrated access patterns.

12 Assumptions Constraints and Resolved Decisions

Assumptions

- The first release uses one seeded Hair Fix Salon location and one salon time zone.
- Each appointment contains one service and one assigned staff member.
- Each account has one primary role in version 1.0.
- Customers may change only their own future Scheduled or Confirmed appointments before the 24-hour cutoff.
- Staff assignments, schedules, service durations, and service prices are maintained before booking tests.

Constraints

- The database will use MySQL and fictional sample data.
- The project records payments but does not connect to a payment processor.
- The project will not store plaintext passwords or sensitive payment credentials.
- A production website, automated notifications, payroll, inventory, and retail sales are outside scope.
- Authentication may be demonstrated with account, hash, role, and privilege data instead of a production login application.
- Multi-location workflows are deferred until a later version.

Resolved Design Decisions

Decision	Version 1.0 choice
Location	One seeded location; no customer selection or multi-location administration in version 1.0.
Roles	One primary role per account.
Names	FirstName and LastName are stored once in UserAccount.
Rescheduling	Update the same Appointment row, preserve prior values in AppointmentHistory, and retain linked payment transactions.
Customer cutoff	24 hours before appointment start.
Payments	Zero or more Charge or Refund transactions may be recorded for any appointment status. Amount is positive; type determines revenue direction.
Revenue	Succeeded charges minus succeeded refunds.
History	Append-only and unavailable to ordinary update or delete operations.

13 Acceptance Criteria and Test Cases

The team will capture the input, actual result, and supporting SQL query or screenshot for each test. The test set includes successful operations, invalid inputs, authorization failures, boundary conditions, and transaction rollback.

Test ID	Requirement	Test type	Test case	Expected outcome
TC-01	FR-01 BR-01	Negative	Insert a second account with an existing email.	Insert fails because the email must be unique.
TC-02	FR-07 FR-08	Positive	Book an active qualified staff member for an open future time.	Appointment and one booking-history row are created with price and duration snapshots.
TC-03	FR-09 BR-08	Negative	Attempt a partially overlapping booking for the same staff member.	The controlled operation rejects the request and creates no rows.
TC-04	FR-11 FR-12	Authorization	Use a Customer account to mark an appointment No-Show.	The status change is rejected.

Test ID	Requirement	Test type	Test case	Expected outcome
TC-05	FR-19	Invalid workflow	Change a Completed appointment back to Scheduled.	The transition is rejected and no misleading history event is created.
TC-06	FR-20 BR-24	Resiliency	Force history insertion or validation to fail during booking.	The full transaction rolls back; no appointment or partial history remains.
TC-07	FR-18	Historical integrity	Change a service's current price and duration after booking.	The existing appointment's BookedPrice and BookedDurationMinutes remain unchanged.
TC-08	FR-24 BR-21	Audit protection	Attempt to update or delete an AppointmentHistory row through the ordinary application role.	The operation is rejected or the table is inaccessible for that action.
TC-09	FR-10	Positive boundary	Reschedule an owned Scheduled appointment more than 24 hours before start.	The same appointment is updated and prior values are preserved in history.
TC-10	FR-10 BR-10	Negative boundary	Customer attempts to reschedule less than 24 hours before start.	The operation is rejected and the appointment remains unchanged.
TC-11	FR-06 BR-15	Precedence	Add a blocked period inside recurring or date-specific availability.	The blocked interval is not returned as available.
TC-12	FR-13 FR-25	Positive	Assigned staff records a positive succeeded Charge with an allowed method.	One payment row is created with the recorder and timestamp.
TC-13	FR-25	Authorization	Customer attempts to create or edit a payment transaction.	The operation is rejected.
TC-14	BR-18 RQ-04	Report	Run revenue with succeeded charges, a succeeded refund, and failed or pending transactions.	Revenue equals succeeded charges minus the succeeded refund; failed and pending rows are excluded.
TC-15	FR-23	Deactivation	Deactivate a service after it has historical appointments, then attempt a new booking.	The new booking is rejected and historical appointments remain.
TC-16	FR-16	Query	Run the daily schedule for one date and staff member.	Only matching appointments appear in start-time order.
TC-17	FR-22 BR-16	Negative	Insert a schedule whose end time is not later than its start time.	The record is rejected.
TC-18	FR-03 BR-25	Negative	Insert a Customer profile without a phone number or with an invalid contact preference.	The profile insert is rejected.
TC-19	FR-10 BR-26	Relationship	Reschedule an appointment that already has a payment transaction.	The same AppointmentID is retained and the payment remains linked without automatic modification.
TC-20	FR-13 BR-27	Final status payment	Record an authorized Refund for a Cancelled appointment and a Charge for a No-Show appointment.	Both valid transactions are recorded without changing either appointment status.

14 Requirement to Design Traceability Matrix

Requirement	Schema objects	Enforcement	Test case	Demonstration evidence
FR-01 to FR-03	UserAccount, Role, Customer	Primary keys, unique email, required phone, contact preference, role foreign key, protected updates	TC-01 TC-18	Account inserts, duplicate-email rejection, customer validation, role query
FR-04 to FR-06	Staff, Service, StaffService, StaffSchedule	Active flags, junction table, schedule validation and precedence	TC-11 TC-17	Qualification rows and availability results
FR-07 to FR-09	Appointment, StaffSchedule, StaffService	Availability query and overlap procedure	TC-02 TC-03	Open-time result, successful booking, rejected conflict
FR-10	Appointment, AppointmentHistory, PaymentTransaction	Ownership, status, cutoff, controlled reschedule, and retained payment link	TC-09 TC-10 TC-19	Before and after appointment, history, and payment queries
FR-11 FR-12 FR-19 FR-21	Appointment, AppointmentStatus, AppointmentHistory	Transition procedure, role validation, required reason	TC-04 TC-05	Authorized change and rejected unauthorized or invalid changes

Requirement	Schema objects	Enforcement	Test case	Demonstration evidence
FR-13 FR-25	PaymentTransaction, PaymentMethod, Appointment	Foreign keys, checks, role, assignment, and final-status payment validation	TC-12 TC-13 TC-20	Successful payments, final-status transactions, and rejected customer operation
FR-14	UserAccount, Appointment, PaymentTransaction	Ownership and assignment filters	Authorization query set	Customer, staff, and administrator result comparisons
FR-15	Operational tables	Administrator procedures and restricted table privileges	Administrative maintenance tests	Seed and maintenance statements
FR-16 FR-17	Appointment, PaymentTransaction, Service, Staff	Views or report queries and documented indexes	TC-14 TC-16	Query results and EXPLAIN evidence
FR-18	Appointment, Service	BookedPrice and BookedDurationMinutes snapshots	TC-07	Service edit followed by unchanged appointment query
FR-20	Appointment, AppointmentHistory	One database transaction with rollback	TC-06	Failure injection and zero partial rows
FR-22	StaffSchedule	Time-range checks and controlled schedule procedure	TC-17	Rejected invalid schedule
FR-23	Service, Staff, Location, Appointment	Active-status validation without historical deletion	TC-15	Rejected booking and retained history
FR-24	AppointmentHistory	Append-only privileges and optional protective trigger	TC-08	Rejected update and delete statements